

# Taller 5

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## Configuración

```
library(DBI)
library(RSQLite)
library(dplyr)
library(knitr)
library(kableExtra)
```

## Conexión a la base de datos

```
ruta_db <- "chinook.db"

if (file.exists(ruta_db)) {
  file.remove(ruta_db)
}

url_chinook <- "https://raw.githubusercontent.com/lerocha/chinook-database/master/ChinookData"

download.file(
  url_chinook,
  destfile = ruta_db,
  mode = "wb"
)

con <- dbConnect(
  RSQLite::SQLite(),
  ruta_db
)
```

```
dbListTables(con)
```

```
[1] "Album"          "Artist"          "Customer"         "Employee"
[5] "Genre"          "Invoice"          "InvoiceLine"      "MediaType"
[9] "Playlist"       "PlaylistTrack"   "Track"
```

## Función para mostrar las tablas

Como el documento se renderiza a PDF (motor LaTeX), las tablas deben generarse con `format = "latex"` y `latex_options`, no con `format = "html"` y `bootstrap_options` (esas opciones solo aplican a salida HTML y por eso las tablas no aparecían).

```
mostrar_tabla <- function(datos) {

  datos |>
    kable(
      format = "latex",
      digits = 2,
      row.names = FALSE,
      align = "c",
      booktabs = TRUE,
      longtable = TRUE
    ) |>
    kable_styling(
      latex_options = c(
        "striped",
        "hold_position",
        "repeat_header"
      ),
      full_width = FALSE,
      position = "center",
      font_size = 9
    )
}
```

## Punto 1

Usando las tablas Customer, Employee, Invoice, InvoiceLine, Track y Album de Chinook.

## Punto 1.a

Cada cliente tiene un empleado de soporte asignado (`Customer.SupportRepId` referencia a `Employee.EmployeeId`), y cada empleado puede tener un jefe (`Employee.ReportsTo`, que también referencia a `Employee.EmployeeId`). Se combina `Customer` con `Employee` mediante `INNER JOIN` y luego una segunda vez la tabla `Employee` (con alias distinto, es decir un `SELF JOIN`) para mostrar: nombre del cliente, nombre del empleado de soporte y nombre del jefe de ese empleado. Se usa `COALESCE` para mostrar 'Sin jefe' cuando el empleado no tenga uno registrado.

```
Punto1a <- dbGetQuery(con, "
SELECT
  Customer.FirstName || ' ' || Customer.LastName AS Cliente,
  Employee.FirstName || ' ' || Employee.LastName AS Empleado_Soporte,
  COALESCE(
    jefe.FirstName || ' ' || jefe.LastName,
    'Sin jefe'
  ) AS Jefe_Empleado
FROM Customer
INNER JOIN Employee
  ON Customer.SupportRepId = Employee.EmployeeId
LEFT JOIN Employee jefe
  ON Employee.ReportsTo = jefe.EmployeeId;
")

mostrar_tabla(Punto1a)
```

| Cliente               | Empleado_Soporte | Jefe_Empleado |
|-----------------------|------------------|---------------|
| Luís Gonçalves        | Jane Peacock     | Nancy Edwards |
| Leonie Köhler         | Steve Johnson    | Nancy Edwards |
| François Tremblay     | Jane Peacock     | Nancy Edwards |
| Bjørn Hansen          | Margaret Park    | Nancy Edwards |
| František Wichterlová | Margaret Park    | Nancy Edwards |
| Helena Holý           | Steve Johnson    | Nancy Edwards |
| Astrid Gruber         | Steve Johnson    | Nancy Edwards |
| Daan Peeters          | Margaret Park    | Nancy Edwards |
| Kara Nielsen          | Margaret Park    | Nancy Edwards |
| Eduardo Martins       | Margaret Park    | Nancy Edwards |
| Alexandre Rocha       | Steve Johnson    | Nancy Edwards |
| Roberto Almeida       | Jane Peacock     | Nancy Edwards |
| Fernanda Ramos        | Margaret Park    | Nancy Edwards |
| Mark Philips          | Steve Johnson    | Nancy Edwards |
| Jennifer Peterson     | Jane Peacock     | Nancy Edwards |

(continued)

| Cliente               | Empleado_Soporte | Jefe_Empleado |
|-----------------------|------------------|---------------|
| Frank Harris          | Margaret Park    | Nancy Edwards |
| Jack Smith            | Steve Johnson    | Nancy Edwards |
| Michelle Brooks       | Jane Peacock     | Nancy Edwards |
| Tim Goyer             | Jane Peacock     | Nancy Edwards |
| Dan Miller            | Margaret Park    | Nancy Edwards |
| Kathy Chase           | Steve Johnson    | Nancy Edwards |
| Heather Leacock       | Margaret Park    | Nancy Edwards |
| John Gordon           | Margaret Park    | Nancy Edwards |
| Frank Ralston         | Jane Peacock     | Nancy Edwards |
| Victor Stevens        | Steve Johnson    | Nancy Edwards |
| Richard Cunningham    | Margaret Park    | Nancy Edwards |
| Patrick Gray          | Margaret Park    | Nancy Edwards |
| Julia Barnett         | Steve Johnson    | Nancy Edwards |
| Robert Brown          | Jane Peacock     | Nancy Edwards |
| Edward Francis        | Jane Peacock     | Nancy Edwards |
| Martha Silk           | Steve Johnson    | Nancy Edwards |
| Aaron Mitchell        | Margaret Park    | Nancy Edwards |
| Ellie Sullivan        | Jane Peacock     | Nancy Edwards |
| João Fernandes        | Margaret Park    | Nancy Edwards |
| Madalena Sampaio      | Margaret Park    | Nancy Edwards |
| Hannah Schneider      | Steve Johnson    | Nancy Edwards |
| Fynn Zimmermann       | Jane Peacock     | Nancy Edwards |
| Niklas Schröder       | Jane Peacock     | Nancy Edwards |
| Camille Bernard       | Margaret Park    | Nancy Edwards |
| Dominique Lefebvre    | Margaret Park    | Nancy Edwards |
| Marc Dubois           | Steve Johnson    | Nancy Edwards |
| Wyatt Girard          | Jane Peacock     | Nancy Edwards |
| Isabelle Mercier      | Jane Peacock     | Nancy Edwards |
| Terhi Hämäläinen      | Jane Peacock     | Nancy Edwards |
| Ladislav Kovács       | Jane Peacock     | Nancy Edwards |
| Hugh O'Reilly         | Jane Peacock     | Nancy Edwards |
| Lucas Mancini         | Steve Johnson    | Nancy Edwards |
| Johannes Van der Berg | Steve Johnson    | Nancy Edwards |
| Stanisław Wójcik      | Margaret Park    | Nancy Edwards |
| Enrique Muñoz         | Steve Johnson    | Nancy Edwards |
| Joakim Johansson      | Steve Johnson    | Nancy Edwards |
| Emma Jones            | Jane Peacock     | Nancy Edwards |
| Phil Hughes           | Jane Peacock     | Nancy Edwards |
| Steve Murray          | Steve Johnson    | Nancy Edwards |
| Mark Taylor           | Margaret Park    | Nancy Edwards |
| Diego Gutiérrez       | Margaret Park    | Nancy Edwards |
| Luis Rojas            | Steve Johnson    | Nancy Edwards |
| Manoj Pareek          | Jane Peacock     | Nancy Edwards |
| Puja Srivastava       | Jane Peacock     | Nancy Edwards |

## Punto 1.b

Con una subconsulta correlacionada, se buscan los clientes cuyo gasto total (`SUM(Invoice.Total)`) sea mayor al gasto promedio de los demás clientes atendidos por el mismo empleado de soporte (no el promedio general, ni el del país: el de su propio representante).

```
Punto1b <- dbGetQuery(con, "
SELECT
  Customer.CustomerId,
  Customer.FirstName || ' ' || Customer.LastName AS Cliente,
  SUM(Invoice.Total) AS total_gastado
FROM Customer
JOIN Invoice
  ON Customer.CustomerId = Invoice.CustomerId
GROUP BY
  Customer.CustomerId,
  Customer.FirstName,
  Customer.LastName,
  Customer.SupportRepId
HAVING SUM(Invoice.Total) > (
  SELECT AVG(GastoOtros)
  FROM (
    SELECT
      SUM(FacturaOtros.Total) AS GastoOtros
    FROM Customer AS ClienteOtros
    JOIN Invoice AS FacturaOtros
      ON ClienteOtros.CustomerId = FacturaOtros.CustomerId
    WHERE ClienteOtros.SupportRepId = Customer.SupportRepId
      AND ClienteOtros.CustomerId <> Customer.CustomerId
    GROUP BY ClienteOtros.CustomerId
  )
);
")

mostrar_tabla(Punto1b)
```

| CustomerId | Cliente          | total_gastado |
|------------|------------------|---------------|
| 24         | Frank Ralston    | 43.62         |
| 37         | Fynn Zimmermann  | 43.62         |
| 43         | Isabelle Mercier | 40.62         |
| 44         | Terhi Hämäläinen | 41.62         |
| 45         | Ladislav Kovács  | 45.62         |

(continued)

| CustomerId | Cliente               | total_gastado |
|------------|-----------------------|---------------|
| 46         | Hugh O'Reilly         | 45.62         |
| 4          | Bjørn Hansen          | 39.62         |
| 5          | František Wichterlová | 40.62         |
| 20         | Dan Miller            | 39.62         |
| 22         | Heather Leacock       | 39.62         |
| 26         | Richard Cunningham    | 47.62         |
| 34         | João Fernandes        | 39.62         |
| 6          | Helena Holý           | 49.62         |
| 7          | Astrid Gruber         | 42.62         |
| 25         | Victor Stevens        | 42.62         |
| 28         | Julia Barnett         | 43.62         |
| 48         | Johannes Van der Berg | 40.62         |
| 57         | Luis Rojas            | 46.62         |

### Punto 1.c

Usando EXCEPT, se identifican los empleados que nunca han sido asignados como soporte de ningún cliente (aparecen en Employee pero jamás en Customer.SupportRepId).

```
punto1c <- dbGetQuery(con, "  
  SELECT  
    Employee.EmployeeId,  
    Employee.FirstName || ' ' || Employee.LastName AS Empleado,  
    Employee.Title AS Cargo  
  FROM Employee  
  WHERE Employee.EmployeeId IN (  
    SELECT EmployeeId  
    FROM Employee  
  
    EXCEPT  
  
    SELECT SupportRepId  
    FROM Customer  
    WHERE SupportRepId IS NOT NULL  
  );  
")  
  
mostrar_tabla(punto1c)
```

| EmployeeId | Empleado         | Cargo           |
|------------|------------------|-----------------|
| 1          | Andrew Adams     | General Manager |
| 2          | Nancy Edwards    | Sales Manager   |
| 6          | Michael Mitchell | IT Manager      |
| 7          | Robert King      | IT Staff        |
| 8          | Laura Callahan   | IT Staff        |

### Punto 1.d

Con una CTE, se calcula cuántos clientes atiende cada empleado y el ingreso total generado por esos clientes, ordenando de mayor a menor ingreso.

```
punto1d <- dbGetQuery(con, "
  WITH ResumenSoporte AS (
    SELECT
      e.EmployeeId,
      e.FirstName || ' ' || e.LastName AS Empleado,
      COUNT(DISTINCT c.CustomerId) AS Cantidad_Clientes,
      COALESCE(SUM(i.Total), 0) AS Ingreso_Total
    FROM Employee e
    LEFT JOIN Customer c
      ON e.EmployeeId = c.SupportRepId
    LEFT JOIN Invoice i
      ON c.CustomerId = i.CustomerId
    GROUP BY
      e.EmployeeId,
      e.FirstName,
      e.LastName
  )

  SELECT
    Empleado,
    Cantidad_Clientes,
    Ingreso_Total
  FROM ResumenSoporte
  ORDER BY Ingreso_Total DESC;
")

mostrar_tabla(punto1d)
```

| Empleado         | Cantidad_Clientes | Ingreso_Total |
|------------------|-------------------|---------------|
| Jane Peacock     | 21                | 833.04        |
| Margaret Park    | 20                | 775.40        |
| Steve Johnson    | 18                | 720.16        |
| Andrew Adams     | 0                 | 0.00          |
| Nancy Edwards    | 0                 | 0.00          |
| Michael Mitchell | 0                 | 0.00          |
| Robert King      | 0                 | 0.00          |
| Laura Callahan   | 0                 | 0.00          |

**Interpretación:** el empleado con mayor ingreso generado es Jane Peacock, mientras que el que atiende más clientes es Jane Peacock. En este caso sí coinciden: el empleado que atiende más clientes es también el que genera más ingresos.

## Punto 2

Usando las tablas Invoice, InvoiceLine, Track y Genre.

### Punto 2.a

Se calcula el total de ventas por año y mes (SUBSTR(InvoiceDate, 1, 7)). Con LAG y LEAD en la misma fila se agregan tanto las ventas del mes anterior como las del mes siguiente, y se calcula la variación porcentual (no absoluta) respecto al mes anterior.

```
punto2a <- dbGetQuery(con, "
  WITH VentasMensuales AS (
    SELECT
      SUBSTR(InvoiceDate, 1, 7) AS AnioMes,
      SUM(Total) AS Ventas
    FROM Invoice
    GROUP BY SUBSTR(InvoiceDate, 1, 7)
  )

  SELECT
    AnioMes,
    Ventas,
    LAG(Ventas) OVER (
      ORDER BY AnioMes
    ) AS VentasMesAnterior,
    LEAD(Ventas) OVER (
```

```

ORDER BY AnioMes
) AS VentasMesSiguiente,
ROUND(
(
(Ventas - LAG(Ventas) OVER (ORDER BY AnioMes))
/ LAG(Ventas) OVER (ORDER BY AnioMes)
) * 100,
2
) || '%' AS VariacionPorcentual
FROM VentasMensuales;
")
mostrar_tabla(punto2a)

```

| AnioMes | Ventas | VentasMesAnterior | VentasMesSiguiente | VariacionPorcentual |
|---------|--------|-------------------|--------------------|---------------------|
| 2021-01 | 35.64  | NA                | 37.62              | NA                  |
| 2021-02 | 37.62  | 35.64             | 37.62              | 5.56%               |
| 2021-03 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2021-04 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2021-05 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2021-06 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2021-07 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2021-08 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2021-09 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2021-10 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2021-11 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2021-12 | 37.62  | 37.62             | 52.62              | 0.0%                |
| 2022-01 | 52.62  | 37.62             | 46.62              | 39.87%              |
| 2022-02 | 46.62  | 52.62             | 44.62              | -11.4%              |
| 2022-03 | 44.62  | 46.62             | 37.62              | -4.29%              |
| 2022-04 | 37.62  | 44.62             | 37.62              | -15.69%             |
| 2022-05 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2022-06 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2022-07 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2022-08 | 37.62  | 37.62             | 36.63              | 0.0%                |
| 2022-09 | 36.63  | 37.62             | 37.62              | -2.63%              |
| 2022-10 | 37.62  | 36.63             | 37.62              | 2.7%                |
| 2022-11 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2022-12 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2023-01 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2023-02 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2023-03 | 37.62  | 37.62             | 51.62              | 0.0%                |
| 2023-04 | 51.62  | 37.62             | 42.62              | 37.21%              |
| 2023-05 | 42.62  | 51.62             | 50.62              | -17.44%             |
| 2023-06 | 50.62  | 42.62             | 37.62              | 18.77%              |

(continued)

| AnioMes | Ventas | VentasMesAnterior | VentasMesSiguiente | VariacionPorcentual |
|---------|--------|-------------------|--------------------|---------------------|
| 2023-07 | 37.62  | 50.62             | 37.62              | -25.68%             |
| 2023-08 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2023-09 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2023-10 | 37.62  | 37.62             | 23.76              | 0.0%                |
| 2023-11 | 23.76  | 37.62             | 37.62              | -36.84%             |
| 2023-12 | 37.62  | 23.76             | 37.62              | 58.33%              |
| 2024-01 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2024-02 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2024-03 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2024-04 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2024-05 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2024-06 | 37.62  | 37.62             | 39.62              | 0.0%                |
| 2024-07 | 39.62  | 37.62             | 47.62              | 5.32%               |
| 2024-08 | 47.62  | 39.62             | 46.71              | 20.19%              |
| 2024-09 | 46.71  | 47.62             | 42.62              | -1.91%              |
| 2024-10 | 42.62  | 46.71             | 37.62              | -8.76%              |
| 2024-11 | 37.62  | 42.62             | 37.62              | -11.73%             |
| 2024-12 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2025-01 | 37.62  | 37.62             | 27.72              | 0.0%                |
| 2025-02 | 27.72  | 37.62             | 37.62              | -26.32%             |
| 2025-03 | 37.62  | 27.72             | 33.66              | 35.71%              |
| 2025-04 | 33.66  | 37.62             | 37.62              | -10.53%             |
| 2025-05 | 37.62  | 33.66             | 37.62              | 11.76%              |
| 2025-06 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2025-07 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2025-08 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2025-09 | 37.62  | 37.62             | 37.62              | 0.0%                |
| 2025-10 | 37.62  | 37.62             | 49.62              | 0.0%                |
| 2025-11 | 49.62  | 37.62             | 38.62              | 31.9%               |
| 2025-12 | 38.62  | 49.62             | NA                 | -22.17%             |

## Punto 2.b

Se construye una CTE con las ventas por año y mes, y a partir de ella se generan dos rankings con `RANK() OVER (PARTITION BY año ORDER BY ventas ...)`: uno para identificar el mes de mayor venta de cada año y otro para el de menor venta. Se muestra en una sola tabla final, por año, el mes top y el mes más bajo.

```
punto2b <- dbGetQuery(con, "  
  WITH VentasMensuales AS (  
    SELECT  
      SUBSTR(InvoiceDate, 1, 4) AS Anio,
```

```

        SUBSTR(InvoiceDate, 1, 7) AS AnioMes,
        SUM(Total) AS Ventas
    FROM Invoice
    GROUP BY SUBSTR(InvoiceDate, 1, 7)
),

Rankings AS (
    SELECT
        Anio,
        AnioMes,
        Ventas,
        RANK() OVER (
            PARTITION BY Anio
            ORDER BY Ventas DESC
        ) AS RankTop,
        RANK() OVER (
            PARTITION BY Anio
            ORDER BY Ventas ASC
        ) AS RankBajo
    FROM VentasMensuales
)

SELECT
    r1.Anio,
    r1.AnioMes AS MesTop,
    r1.Ventas AS VentasTop,
    r2.AnioMes AS MesBajo,
    r2.Ventas AS VentasBajo
FROM Rankings r1
JOIN Rankings r2
    ON r1.Anio = r2.Anio
WHERE r1.RankTop = 1
    AND r2.RankBajo = 1;
")

```

mostrar\_tabla(punto2b)

| Anio | MesTop  | VentasTop | MesBajo | VentasBajo |
|------|---------|-----------|---------|------------|
| 2021 | 2021-02 | 37.62     | 2021-01 | 35.64      |
| 2021 | 2021-03 | 37.62     | 2021-01 | 35.64      |
| 2021 | 2021-04 | 37.62     | 2021-01 | 35.64      |
| 2021 | 2021-05 | 37.62     | 2021-01 | 35.64      |

(continued)

| Anio | MesTop  | VentasTop | MesBajo | VentasBajo |
|------|---------|-----------|---------|------------|
| 2021 | 2021-06 | 37.62     | 2021-01 | 35.64      |
| 2021 | 2021-07 | 37.62     | 2021-01 | 35.64      |
| 2021 | 2021-08 | 37.62     | 2021-01 | 35.64      |
| 2021 | 2021-09 | 37.62     | 2021-01 | 35.64      |
| 2021 | 2021-10 | 37.62     | 2021-01 | 35.64      |
| 2021 | 2021-11 | 37.62     | 2021-01 | 35.64      |
| 2021 | 2021-12 | 37.62     | 2021-01 | 35.64      |
| 2022 | 2022-01 | 52.62     | 2022-09 | 36.63      |
| 2023 | 2023-04 | 51.62     | 2023-11 | 23.76      |
| 2024 | 2024-08 | 47.62     | 2024-01 | 37.62      |
| 2024 | 2024-08 | 47.62     | 2024-02 | 37.62      |
| 2024 | 2024-08 | 47.62     | 2024-03 | 37.62      |
| 2024 | 2024-08 | 47.62     | 2024-04 | 37.62      |
| 2024 | 2024-08 | 47.62     | 2024-05 | 37.62      |
| 2024 | 2024-08 | 47.62     | 2024-06 | 37.62      |
| 2024 | 2024-08 | 47.62     | 2024-11 | 37.62      |
| 2024 | 2024-08 | 47.62     | 2024-12 | 37.62      |
| 2025 | 2025-11 | 49.62     | 2025-02 | 27.72      |

## Punto 2.c

Se encadena una segunda CTE para calcular, por año, cuál fue el género musical con mayores ingresos ( $\text{UnitPrice} \times \text{Quantity}$  vía  $\text{InvoiceLine} \rightarrow \text{Track} \rightarrow \text{Genre}$ ). Se usa `RANK() OVER (PARTITION BY año ORDER BY ingresos DESC)` y se filtra solo la posición 1 de cada año con un `WHERE` sobre el resultado de la CTE.

```
punto2c <- dbGetQuery(con, "  
  WITH GeneroPorAnio AS (  
    SELECT  
      SUBSTR(Invoice.InvoiceDate, 1, 4) AS Anio,  
      Genre.Name AS Genero,  
      SUM(  
        InvoiceLine.UnitPrice * InvoiceLine.Quantity  
      ) AS Ingresos  
    FROM Invoice  
    JOIN InvoiceLine  
      ON Invoice.InvoiceId = InvoiceLine.InvoiceId  
    JOIN Track  
      ON InvoiceLine.TrackId = Track.TrackId  
    JOIN Genre  
      ON Track.GenreId = Genre.GenreId
```

```

GROUP BY
    SUBSTR(Invoice.InvoiceDate, 1, 4),
    Genre.Name
),

RankingGenero AS (
    SELECT
        Anio,
        Genero,
        Ingresos,
        RANK() OVER (
            PARTITION BY Anio
            ORDER BY Ingresos DESC
        ) AS Posicion
    FROM GeneroPorAnio
)

SELECT
    Anio,
    Genero,
    Ingresos
FROM RankingGenero
WHERE Posicion = 1;
")

mostrar_tabla(punto2c)

```

| Anio | Genero | Ingresos |
|------|--------|----------|
| 2021 | Rock   | 178.20   |
| 2022 | Rock   | 155.43   |
| 2023 | Rock   | 156.42   |
| 2024 | Rock   | 162.36   |
| 2025 | Rock   | 174.24   |

## Punto 2.d

Sobre el gasto total por cliente, se usa `NTILE(4)` para dividirlos en cuartiles y luego, con una CTE adicional que agrega por cuartil, se calcula qué porcentaje del ingreso total de la tienda representa el cuartil superior (Q4) frente a los otros tres cuartiles combinados.

```

punto2d <- dbGetQuery(con, "
WITH GastoCliente AS (
  SELECT
    CustomerId,
    SUM(Total) AS GastoTotal,
    NTILE(4) OVER (
      ORDER BY SUM(Total) ASC
    ) AS Cuartil
  FROM Invoice
  GROUP BY CustomerId
),

VentasPorCuartil AS (
  SELECT
    Cuartil,
    SUM(GastoTotal) AS TotalCuartil
  FROM GastoCliente
  GROUP BY Cuartil
),

Comparacion AS (
  SELECT
    SUM(
      CASE
        WHEN Cuartil = 4
        THEN TotalCuartil
        ELSE 0
      END
    ) AS Ingreso_Q4,

    SUM(
      CASE
        WHEN Cuartil < 4
        THEN TotalCuartil
        ELSE 0
      END
    ) AS Ingreso_Otros,

    SUM(TotalCuartil) AS Ingreso_Total
  FROM VentasPorCuartil
)

```

```

SELECT
    Ingreso_Q4,
    Ingreso_Otros,
    Ingreso_Total,
    ROUND(
        (Ingreso_Q4 / Ingreso_Total) * 100,
        2
    ) || '%' AS Porcentaje_Q4,
    ROUND(
        (Ingreso_Otros / Ingreso_Total) * 100,
        2
    ) || '%' AS Porcentaje_Otros
FROM Comparacion;
")

```

mostrar\_tabla(punto2d)

| Ingreso_Q4 | Ingreso_Otros | Ingreso_Total | Porcentaje_Q4 | Porcentaje_Otros |
|------------|---------------|---------------|---------------|------------------|
| 614.68     | 1713.92       | 2328.6        | 26.4%         | 73.6%            |

## Cierre

dbDisconnect(con)